

Manwe Castro

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Education

Bachelor of Science in Mechanical Engineering

Arizona State University | GPA: 3.61

Tempe, AZ

May 2028

Relevant Coursework: Mechanics of Materials, Structure and Properties of Materials, Thermodynamics, Dynamics, Circuits

Project Experience

Mental Health Robotics System for Isolated Environments

Tempe, AZ

Design and Prototyping Lead

August 2024 – December 2024

- Programmed an Arduino Uno to process ultrasonic and photoresistive sensor inputs, controlling drive motors and a servo actuator to enable autonomous user tracking and automated medical dispensing, achieving 95%+ proximity detection reliability, repeatable 15 cm stopping distance, and drawer activation below a 25 Lux light value threshold.
- Modeled a 1.8 cubic foot robotic enclosure in SolidWorks and fabricated the structure from laser-cut balsa and plywood, integrating sensors, motors, display hardware, and battery components into a prototype build, finishing 40% under the \$75 budget while maintaining lightweight structural rigidity.
- Executed formal acceptance testing across functional requirements, diagnosing and resolving an electrical power delivery fault to pass sensor, motor, and dispensing tests, validating semi-autonomous navigation and automated medical access.

VR Rapid Prototyping Application

Tempe, AZ

Prototyping Lead | Modeling Team

March 2025 – December 2025

- Engineered a virtual reality prototyping environment for Meta Quest headsets using Unity and C#, integrating over 100 interactive 3D assets with simulated physics and collision mechanics to support rapid design across 30+ headsets
- Directed a cross-functional modeling team by establishing standardized Unity workflows and resolving engine-level rendering blockers, optimizing frame rates to mitigate VR motion sickness
- Collaborated with faculty to integrate a 2-week curriculum into the experience and developed in-game analytics scripts to track completion time, errors, and hints, creating a data-driven feedback loop for instructors.
- Authored Gantt charts, risk matrices, and standardized technical reports for a 12-person, multi-semester project, delivering a transition package that enabled future teams to continue development, testing, and classroom deployment.

Shock Absorber Assembly

August 2025 – December 2025

- Designed a fully constrained automotive shock absorber assembly in SolidWorks by modeling custom components, integrating McMaster-Carr hardware, and amending dimensions to eliminate physical interferences for dynamic suspension applications.
- Generated fully dimensioned 2D engineering drawings following ASME Y14.5 standards, detailing complex features such as shaft thread specifications, keyways, and spring parameters to provide precise fabrication instructions.
- Authored a comprehensive Bill of Materials and exploded assembly drawing to document custom and standard hardware, establishing a standardized technical reference to streamline part procurement.

Professional Experience

Institute for Digital Inclusion Acceleration

Chandler, AZ

Digital Navigator

January 2025 – Present

- Instruct over 300 weekly visitors in hardware and software fundamentals, facilitating thousands of hands-on technology interactions across diverse age groups.
- Develop and deliver structured curriculum for nine distinct software platforms, including Swift, TinkerCAD, and SpheroEDU, standardizing the instructional process for the center.
- Operate and maintain Flashforge 3D printers using Orca Slicer, troubleshooting hardware failures and optimizing print parameters to ensure consistent output quality.

Leadership and Campus Involvement

Society of Hispanic Professional Engineers

Tempe, AZ

Outreach Director

March 2026 – Present

- Direct chapter outreach by partnering with local schools and community organizations, prioritizing schools with large Hispanic student populations.
- Plan and coordinate outreach and professional development events of up to 50 attendees including Noche de Ciencias and Engineers Week.
- Manage a 4-member student events committee, 15+ volunteers, and a \$500 annual budget while designing 4 hands-on technical workshops throughout the semester that build practical engineering skills for undergraduate members.

Skills

CAD & Design: SolidWorks, Fusion 360, 2D Drafting, MATLAB

Fabrication & Hands-On: FDM 3D Printing, Circuit Wiring, Laser Cutting, Orca Slicer, Automotive Repair

Programming & Controls: Python, Arduino IDE (C++), Unity Game Engine, Git, Sensor Integration

Personal: Violin, Guitar, Sewing, Piano, Running, Jarana, Requinto, Boxing, Rockclimbing, Photography, Salsa